

Hasnat Rasool

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SUMMARY

AI/ML Software Engineer with 2+ years building agentic AI systems, RAG pipelines, and full-stack platforms (Next.js, NestJS, Flask, Angular). Specialized in LLM cost/latency optimization (LangGraph, LangSmith), multi-tenant SaaS architecture, and high-throughput data systems processing 4M+ records.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, HTML/CSS

AI/ML & Agentic Systems: LangGraph, LangSmith, LangChain, RAG, Ollama, ChromaDB, VectorDB, OpenAI, Gemini, Groq, Serper

Frameworks: Next.js, NestJS, React.js, Angular, Node.js, Flask, .NET, n8n, Electron.js

Databases: MongoDB, MS SQL Server, PostgreSQL

Tools & Cloud: Git/GitHub, AWS, Docker, REST APIs, GitHub API, Hatchet, Workflow Automation

EDUCATION

Bachelor of Science in Computer Science

National University of Computer and Emerging Sciences (FAST)

2020 – 2024

Islamabad, Pakistan

EXPERIENCE

Software Engineer - AI/ML

Techverx

May 2026 – Present

Lahore, Pakistan

- Migrated an agentic AI workflow (OpenAI GPT, Serper search integration) from n8n to LangGraph, redesigning orchestration logic and cutting average token consumption per run from 250K to 15K–20K (92%+ reduction), with full execution tracing via LangSmith.
- Architected and built a full-stack PMO engine (Next.js, NestJS) for dynamic project, task, and resource management, fully configurable with no hardcoded logic, supporting multi-tenant architecture and role-based access control (RBAC).
- Integrated GitHub API and Hatchet workflow orchestration to automate cross-project task syncing and background job execution.

Full-Stack Software Developer

Angularspring, Powersoft19

Jan 2025 – Apr 2026

Lahore, Pakistan

- Engineered a 3-Tier AI Waterfall (Gemini 2.5 Flash, Groq, local Qwen2.5) for a Flask microservice processing 4M+ products, reducing API latency by 90%+ while guaranteeing 100% offline failover uptime.
- Built an LLM data extraction engine with strict Pydantic schema enforcement, auto-healing retries, categorical unpacking, and anti-hallucination gates, eliminating data fabrication.
- Developed a high-speed classification engine using C++ RapidFuzz for MS SQL matching, ensuring thread-safe WSGI concurrency via OS-level semaphores and sub-millisecond retrieval via SHA-256 semantic MongoDB caching.
- Built RAG chatbots (LangChain, Ollama/Llama 3.2, ChromaDB) for context-aware technical assistance, and a cross-platform sync engine (n8n, Evolution API) integrating WhatsApp and Smartsheet into executive dashboards.

Earlier: Web Development Intern @ CyberSoft Vantage; Teaching Assistant @ FAST-NUCES; Event Manager @ TEDxFASTIslamabad

PROJECTS

TrustIsMust | MERN, Python, Machine Learning, AWS

- Built a full-stack fake review detection system using Naive Bayes classification and PyABSA aspect-based sentiment analysis, deployed on AWS (EC2, S3, RDS).

CSV Time Series Visualization Tool | Angular, TypeScript, ECharts WebGL

- Built a high-performance Angular app visualizing 1M+ data points with custom timestamp parsing and streaming CSV parsing (PapaParse) for 500MB+ datasets.

CERTIFICATIONS

- AWS Academy Graduate: AWS Academy Microservices and CI/CD Pipeline Builder
- freeCodeCamp: JavaScript Algorithms and Data Structures
- W3 School: HTML and CSS
- Datacamp: Introduction to Python